Technical Brief: Hydroengineering and Geologic Intelligence Modules

A Governed Landscape-Material Profile for Classified Earthworks

The geologic intelligence module treats sampling, testing, transformation, deployment, and monitoring as linked data events.

Author
Adam Ableman
Constitutional Computing
Contact: ableman.research@gmail.com

Abstract

The Hydroengineering and Geologic Intelligence Modules extend Classified Earthworks from material separation and transformation into a governed landscape-intervention architecture. The hydroengineering module asks how classified mineral, organic, ceramic, compost, biochar, and aggregate fractions might participate in site-specific water management. The geologic intelligence module treats sampling, testing, transformation, deployment, and monitoring as linked data events.

The central object is not "improved dirt" but a bounded suitability relation among a material batch, intended use, site, design, evidence, authority, and time. A material passport records provenance, tests, uncertainty, restrictions, decisions, and later observations. It does not itself certify safety or authorize deployment. Likewise, general patterns such as slowing runoff or increasing infiltration are design possibilities, not universal prescriptions: infiltration can mobilize contaminants, destabilize slopes, damage structures, alter groundwater, or displace flooding.

This brief defines the two modules, their shared ontology, decision gates, passport lifecycle, hydrologic and data-governance constraints, monitoring loop, realization funnel, and evaluation program. It remains a system proposal pending qualified site, laboratory, engineering, legal, community, and environmental review.

Source, Evidence, and Standing

This is an AI-assisted integrated edition derived from one exact author manuscript. Its custody and comparison manifest is
`work/paper-saturation-v1/evidence/`

`hydroengineering-geologic-intelligence-modules/integration-custody-manifest.v0.1.json`, SHA-256 `F7268158A849C67C17E2AFF43ACFCC4684BC60366F70C1ED614844B7C9BBE057`. The source remains unchanged.

The modules, structures, workflow, metrics, and design thesis are author material. Ontology, formal qualification, evidence boundaries, gates, tests, and decisions are AI editorial or synthetic candidates until adopted. No site was assessed, material approved, structure designed, water routed, data released, or external reference verified.

1. System Role

Classified Earthworks proposes that excavated soil, rock, sediment, and biomass should be separated, tested, transformed, and routed according to their properties rather than treated as undifferentiated spoil. This brief adds two coupled profiles:

- **Hydroengineering:** proposes and evaluates landscape interventions that change water timing, path, storage, contact, treatment, and release.
- **Geologic intelligence:** acquires, governs, and revises knowledge about materials, subsurface conditions, site relations, and intervention performance.

The modules are coupled but not identical. Data can exist without authorizing an intervention. A design can be proposed while evidence is insufficient. Monitoring can invalidate an earlier suitability decision.

2. Core Ontology

The architecture distinguishes:

- **site:** a spatial, legal, ecological, hydrologic, and social context;
- **sample:** material collected through a declared sampling method;
- **batch:** a bounded material population with custody and transformation history;
- **test result:** an observation produced by a named method, laboratory or instrument, date, and uncertainty;
- **material profile:** reviewed synthesis of applicable observations;
- **intended use:** a function with performance and exclusion criteria;
- **intervention:** a designed change to land, water, material, or access;
- **decision:** an authority-scoped disposition;
- **observation:** later evidence about construction or performance;
- **revision:** a change in profile, permission, restriction, or design.

Suitability is relational:

$$Suitable(b, u, s, d, e, a, t) \in \{admit, pilot, condition, defer, refuse, isolate, unknown\} \quad ,$$

where b is batch, u use, s site, d design, e evidence, a authority, and t time. A batch is never simply "good" or "clean" outside a purpose, threshold, jurisdiction, and evidence window.

3. Material Passport as a Governed Claim Surface

A material passport should bind:

```
MaterialPassport {
  batch_id
  source_site_and_interval
  custody_and_transformation_history
  sampling_plan_and_representativeness
  test_methods_results_units_uncertainty
  material_profile
  candidate_uses
  prohibited_or_deferred_uses
  decision_authority_and_basis
  spatial_temporal_validity
  deployment_links
  monitoring_links
  revision_history
}
```

Passport postures should remain explicit: `UNSAMPLED`, `SAMPLED`, `TESTED`, `CLASSIFIED`,
`CANDIDATE_USE`, `CONDITIONALLY_APPROVED`, `APPROVED_BY_NAMED_AUTHORITY`, `DEPLOYED`,
`MONITORED`, `RESTRICTED`, `QUARANTINED`, `REVOKED`, or `UNKNOWN`.

A passport is not a permit, warranty, geotechnical certification, contaminant clearance, or proof that a heterogeneous batch matches a small sample. Claims remain bounded by sampling design, chain of custody, laboratory scope, detection limits, spatial variability, transformation, age, and later mixing.

4. Hydroengineering Decision Model

The authorial principle is to change water timing, path, contact surfaces, storage, and release rather than merely capture volume. Candidate structures include terraces, swales, rain gardens, infiltration basins, recharge cells, root-zone reservoirs, drainage layers, sediment forebays, contour bands, cistern systems, and gravity-fed conveyance.

These are Configuration Manifold candidates, not recipes. Every intervention should declare at least:

- design storm or event range and overflow path;
- contributing area, inflow, outflow, storage, infiltration, and evapotranspiration assumptions;
- soil, geology, slope, groundwater, utilities, structures, and downstream receptors;
- source-water and material quality;
- erosion, sediment, clogging, settlement, and stability behavior;
- inspection, maintenance, failure, and decommissioning plans;
- affected rights, permissions, communities, and ecosystems.

A basic water-balance constraint is:

$$\Delta S = P + Q_{in} - Q_{out} - ET - R - L,$$

where S is bounded storage, P precipitation, Q_{in} and Q_{out} surface or managed flows, ET evapotranspiration, R deep recharge, and L other modeled losses. Each term is site- and timescale-specific. The equation organizes accounting; it does not replace a calibrated model or design.

5. Layered Material Bodies

The source's layered structures express an important invariant: biological, structural, filtration, storage, and drainage functions should not be collapsed into a homogeneous amended soil. Candidate layers include surface protection, root-zone media, storage media, structural fill, transition zones, drainage, and overflow protection.

However, labels such as "ceramic-rich," "biochar," "compost," or "coarse aggregate" do not establish performance. Properties depend on feedstock, process, grading, durability, chemistry, contaminants, placement, compaction, saturation, freeze-thaw or wet-dry cycles, roots, and maintenance. Each claimed function requires a material specification and evidence appropriate to the intended use.

6. Infiltration and Recharge Boundaries

"Slow, spread, sink" is not universally safe. Water may need to be infiltrated, detained, treated, bypassed, conveyed, or excluded. Infiltration and recharge candidates require site-specific analysis of contaminant mobilization, source-water quality, groundwater depth and use, soil and rock pathways, slope stability, adjacent foundations, utilities, flooding, ecological effects, rights, permits, and long-term maintenance.

Recharge is a stronger claim than infiltration. Infiltration at the surface does not by itself demonstrate beneficial aquifer recharge, recovery, water-quality protection, or absence of off-site effects. The brief therefore uses `RECHARGE_CANDIDATE` until hydrogeologic and regulatory evidence supports a stronger posture.

Qanat-inspired, subsurface, tunnel, cave, cistern, gallery, and groundwater-interception concepts remain conceptual only. They require qualified geotechnical, structural, hydrologic, environmental, archaeological or cultural, occupational-safety, property-rights, and legal review as applicable.

7. Geologic Intelligence and Sampling Discipline

The geologic intelligence module can record location, depth, horizon, parent material, grading, plasticity, density, permeability, water behavior, chemistry, salinity, organic content, nutrients, contaminants, mineral characteristics, biological indicators, and candidate use classes.

The measurement program must also record what it does not know. Required controls include:

- sampling objective and population;
- spatial and depth coverage;
- field and laboratory methods;
- units, calibration, detection limits, and uncertainty;
- duplicates, blanks, controls, and quality flags where applicable;
- custody from collection through transformation;
- heterogeneity and representativeness limits;
- schema and vocabulary version;
- qualified interpretation and review status.

Maps are model products, not the terrain. Resolution, interpolation, age, coordinate reference, uncertainty, and intended use must accompany every suitability or exclusion layer.

8. Data Governance

Subsurface and landscape data can affect liability, land value, development, ecological protection, water and mineral interests, cultural places, and community power. Findability and interoperability do not imply a right to collect, combine, infer, or disclose.

The system should distinguish public, restricted, protected, and prohibited data; define collection and purpose authority; minimize unnecessary precision; record consent or other lawful basis; constrain secondary use; support correction and revocation where applicable; and preserve community, tribal, landholder, worker, and ecological interests.

The Governed Meaning Base can hold typed relations among batches, samples, methods, designs, decisions, evidence, access rules, and revisions. It must preserve provenance and standing rather than presenting every entry as equally true or public.

9. Decision and Effect Gates

A responsible workflow separates:

1. orientation and known-history review;
2. sampling and investigation design;
3. field collection and custody;
4. laboratory and in-situ testing;
5. material and site profile synthesis;
6. alternative hydrologic designs;
7. professional, legal, environmental, and community review;
8. authorization of a bounded pilot or implementation;
9. construction quality evidence;
10. monitoring, maintenance, reclassification, and revision.

No earlier stage silently authorizes a later one. In particular, an AI-generated recommendation, database classification, material passport, or conceptual paper cannot authorize excavation, infiltration, recharge, structural fill, contaminant handling, data release, or subsurface construction.

10. Outcome and Monitoring Model

Performance is multidimensional:

$$Y = (runoff, peak, storage, infiltration, recharge, erosion, \\ quality, stability, ecology, maintenance, equity).$$

Improvement in one component can worsen another. Increased infiltration may reduce surface runoff while increasing contaminant transport or slope risk. Sediment capture may protect downstream water while concentrating material requiring safe handling. Tree growth may improve while structural settlement worsens.

Monitoring should bind baseline, intervention, comparison where feasible, weather and operating conditions, instrument methods, uncertainty, inspection, maintenance, and adverse events. Causal claims require designs capable of distinguishing intervention effects from rainfall variation, vegetation succession, upstream changes, or measurement drift.

11. Failure Modes

Major failure modes include contaminated material routed to sensitive use; nonrepresentative samples; mixed batches retaining an obsolete passport; unstable slopes; blocked or undersized overflow; accelerated downstream flooding; groundwater contamination; salinity accumulation; clogged infiltration surfaces; settlement; organic material in structural zones; unsafe subsurface work; maintenance abandonment; sensitive-data disclosure; authority overreach; community exclusion; and attractive maps that exceed their evidence.

The appropriate response can be refuse, defer, isolate, redesign, restrict, monitor, repair, decommission, or restore. Reuse percentage is not a success metric when it pressures the system to deploy unsuitable material.

12. Realization Funnel

$$
\begin{aligned}
&\mathit{Possibility} \\
&\rightarrow \mathit{Specification} \\
&\rightarrow CM_1 \\
&\rightarrow CM_2 \\
&\rightarrow IC \\
&\rightarrow IR \\
&\rightarrow \mathit{Evidence} \\
&\rightarrow \mathit{Revision}
\end{aligned}
$$

- **Possibility:** imagine classified-material and water-management relationships.
- **Specification:** define ontology, tests, thresholds, authority, data rights, and performance claims.
- **CM1:** enumerate admissible material routes and interventions, including weak, risky, or refusal cases.
- **CM2:** select preferable candidates under site, safety, ecological, social, economic, and maintenance criteria.
- **IC:** bind an Instance Constitution or Instance Configuration for a named site and project.
- **IR:** construct the Instance Realization or Implementation Realization under qualified authority.
- **Evidence:** measure material, construction, hydrologic, geotechnical, ecological, and governance outcomes.
- **Revision:** update passports, designs, restrictions, and system claims.

This brief develops Specification and candidate CM1/CM2 structure. It supplies no site IC, built IR, or field evidence. Realization-stage IR remains distinct from Intermediate Representation in MCIR and compiler contexts.

13. Evaluation Program

Evaluation should test passport custody and representativeness; inter-laboratory and field/laboratory agreement; material-property stability after transformation; routing accuracy; design sensitivity; overflow and failure behavior; infiltration and water-quality effects; slope and settlement response; clogging and maintenance; monitoring coverage; adverse-event detection; access-control and disclosure resistance; community and tribal governance processes where applicable; and the accuracy of all suitability, exclusion, and causal claims.

Two-Module Profile Under Classified Earthworks

Across the completed P1 corpus, the Hydroengineering and Geologic Intelligence Modules form the **two-module landscape-material profile under Classified Earthworks**. Hydroengineering governs proposed changes to water timing, path, contact, storage, treatment, overflow, and release. Geologic intelligence governs sampling, material and site observations, uncertainty, mapping, access, and revision. Neither module independently authorizes material use, excavation, infiltration, recharge, construction, or data release.

- **Adjacent work:** *Classified Earthworks*
  Supplies: Parent material classification and governed routing architecture.
  Module boundary: The modules specialize site-water design and site/material knowledge.
- **Adjacent work:** *Technical Brief: Classified Earthworks System*
  Supplies: Concise platform, mobile processing, passport, and deployment framing.
  Module boundary: This profile adds deeper hydrologic and geologic decision contracts.
- **Adjacent work:** *Governance Witnesses*
  Supplies: Typed laboratory, field, monitoring, inspection, and community observations.
  Module boundary: Witnesses support bounded suitability claims without certifying the site.
- **Adjacent work:** *A Formal Theory of Work Custody*
  Supplies: Custody for samples, batches, designs, decisions, placements, and monitoring duties.
  Module boundary: Custody continuity is required but not sufficient for technical suitability.
- **Adjacent work:** *The Constitutional Observability Platform*
  Supplies: Separation of domain state, presentation, sensors, accessibility, and interpretation.
  Module boundary: It helps prevent maps and dashboards from becoming the terrain or the effect.

A module decision should bind:

$$\begin{aligned} D = (&site, batch, use, design, sampling, tests, hydrology,\\ &geology, uncertainty, rights, authority, data_policy,.\\ &decision, effect, monitoring, decommissioning) \end{aligned}$$

Suitability is relational and temporal. A batch or site is not simply good, clean, infiltrating, or recharge-producing. `ADMIT`,
`PILOT`, `CONDITION`, `DEFER`, `REFUSE`, `ISOLATE`,

and UNKNOWN remain separate. Surface infiltration does not establish beneficial recharge; a material passport does not grant a permit; a map does not remove interpolation uncertainty.

The modules must preserve downstream and off-site effects, extreme-event overflow, contaminant mobilization, slope and structural response, maintenance, adverse-event detection, community and tribal governance, protected data, and decommissioning. Reuse or infiltration targets cannot override a do-not-use or do-not-infiltrate decision.

The universal GSM can type
SPECIALIZES_CLASSIFIED_EARTHWORKS,
CHARACTERIZES_SITE_GEOLOGY, PROPOSES_HYDROLOGIC_INTERVENTION,
SAMPLES_BATCH_AT_SITE,
CONDITIONS_USE_ON_EVIDENCE, REFUSES_INFILTRATION, REMAINS_RECHARGE_CANDIDATE,
MONITORS_DOWNSTREAM_EFFECT, RESTRICTS_SITE_DATA,
and
DECOMMISSIONS_INTERVENTION. These relations preserve the parent/profile structure.

This integration establishes four editorial commitments: retain the work as a two-module profile under Classified Earthworks; adopt the module-decision envelope; reserve recharge, safety, and suitability claims for site-specific evidence and authority; and require monitoring, protected-data, downstream-effect, refusal, and decommissioning semantics.

Protected Site Meaning, Learning Boundaries, and Exact Decision Lineage

The fixed-seed comparison with the Suppression Paradox, governed continual training, project-state lineage, and The Locus of Meaning expands a site decision from a technical record into a governed semantic and evidentiary binding:

$$\begin{aligned} D^* = (&site, batch, use, design, sampling, tests, hydrology,\\ &geology, uncertainty, rights, authority,\\ &protected_constraints, meaning_binding, lineage,\\ &decision, effect, monitoring, revision, decommissioning) \end{aligned}.$$

Protected community, tribal, ecological, safety, location, or infrastructure information may materially constrain a design even when it cannot be disclosed in a public map or general model. An authorized representation can carry the constraint, custodian, scope, review route, and expiry; secrecy alone does not prove the constraint or remove contestability.

Continual learning may update sampling priorities, anomaly detection, forecasting, or maintenance hypotheses from monitoring. It cannot silently weaken a

do-not-infiltrate decision, erase adverse outcomes, or promote a site-wide suitability claim. Project-state lineage should bind sample, passport, model, design, decision, construction, monitoring, and decommissioning generations without treating byte identity as material identity.

The Locus of Meaning supplies a semantic boundary: the meaning of a passport, map symbol, threshold, or suitability label resides in its typed binding among material, site, use, method, jurisdiction, evidence, uptake, and consequence. It is not stored wholly in the document or inferred solely by the reader.

The universal GSM can type
`CARRIES_PROTECTED_SITE_CONSTRAINT`,
`LEARNS_FROM_MONITORING_UNDER_GATE`, `PRESERVES_SAMPLE_TO_DECISION_LINEAGE`,
`BINDS_PASSPORT_MEANING_IN_JURISDICTION`, `REFUSES_SITEWIDE_GENERALIZATION`,
`REVISES_AFTER_ADVERSE_EFFECT`, and
`DECOMMISSIONS_WITH_REMAINDER`.

This randomized review establishes four editorial commitments: adopt the expanded site-decision envelope; create a protected constraint channel with custody and appeal; govern monitoring-driven learning without self-authorizing promotion; and bind passport and map meaning to exact project lineage plus site/use/evidence jurisdiction.

One Physical Decision Braid Across Six Downstream Uses

Six material downstream uses confirm this work as a profile under Classified Earthworks rather than an independent authorization system. The parent paper and technical system brief use it for site-water and geologic specialization. Governed continual training uses monitoring as a source of bounded model proposals while preserving permits, professional judgment, protected constraints, and do-not-use decisions. Project lineage uses the profile to distinguish sample, batch, passport, model, design, placement, monitoring, and decommissioning succession. The Locus of Meaning binds passport and map semantics to site, method, use, jurisdiction, evidence, uptake, and consequence. The Suppression Paradox carries protected constraints without transferring physical evidence or domain authority.

Together they form a physical decision braid:

$$\begin{aligned} D = (&physical_subject, semantic_identity, custody, site,\\ &use, sample, method, model, protected_constraints,\\ &authority, design, decision, effect, monitoring, lineage,\\ &revision, decommissioning, remainder) \end{aligned}.$$

No strand substitutes for another. Byte lineage is not material continuity. A material passport is not a permit. A model update is not professional approval. A coherent map is not the terrain. A protected constraint can remain binding

without public disclosure, but secrecy alone is not evidence and must retain a custodian, scope, review path, and expiry or revision rule.

Every physical effect creates new evidence obligations. Placement, infiltration, overflow, contaminant movement, settlement, slope response, maintenance, community impact, and downstream conditions can defeat an earlier design or suitability claim. Learning and semantic revisions must point to the exact physical generation and cannot overwrite adverse outcomes or refusal history.

The universal GSM can type
`SPECIALIZES_CLASSIFIED_EARTHWORKS_PROFILE`,
`BINDS_PHYSICAL_DECISION_BRAID`, `PRESERVES_MATERIAL_SUCCESSION`,
`LEARNS_FROM_MONITORING_WITHOUT_AUTHORITY`, `CARRIES_PROTECTED_SITE_CONSTRAINT`,
`REVISES_AFTER_PHYSICAL_EFFECT`, and
`DECOMMISSIONS_WITH_RESIDUAL_REMAINDER`.

This reverse-dependency review establishes four editorial commitments: adopt the physical decision braid; keep the work subordinate to the Classified Earthworks identity; require every learning or semantic successor to bind exact physical lineage; and preserve protected constraints, adverse outcomes, refusal history, and decommissioning obligations through revision.

14. Counterevidence and Boundary Tests

The proposal must narrow or fail if: (1) passported batches are materially heterogeneous; (2) samples are not representative; (3) custody is broken; (4) test methods do not fit the use; (5) detection limits exceed relevant thresholds; (6) transformation changes material properties; (7) mixing invalidates prior results; (8) increased infiltration mobilizes contaminants; (9) infiltration destabilizes a slope; (10) storage displaces flooding downstream; (11) overflow is blocked or undersized; (12) sediment traps create a new waste stream; (13) biochar, compost, or ceramic labels substitute for specifications; (14) structural fill contains unsuitable organics; (15) root-zone success masks settlement; (16) recharge is claimed from surface infiltration alone; (17) groundwater or neighboring structures are harmed; (18) salinity or waterlogging accumulates; (19) extreme events exceed assumptions; (20) maintenance is unavailable; (21) maps hide interpolation uncertainty; (22) data is disclosed beyond authority; (23) communities lack meaningful governance; (24) rights or permits block the design; (25) reuse targets override safety; (26) monitoring cannot attribute effects; (27) adverse outcomes appear outside the project boundary; (28) AI classifications are treated as professional approval; (29) external sources or standards do not support attributed claims; (30) lifecycle cost or energy reverses the benefit; (31) the intervention cannot be safely decommissioned; or (32) a do-not-infiltrate decision is not preserved through design and construction.

Water and Geologic Data Do Not Authorize Intervention

Public orientation

The Hydroengineering and Geologic Intelligence Modules join physical material custody to site-specific water decisions. Their central object is a bounded suitability relation among a material batch, intended use, site, design, evidence, authority, and time. The modules can organize investigation and monitoring; they do not turn a general water principle or material passport into permission to alter land or water.

Adversarial tests

1. **Infiltration-benefit attack.** Increased infiltration is assumed beneficial despite contaminant transport, slope instability, structural damage, groundwater effects, saturation, or displaced flooding.
2. **Sample-site attack.** Point observations or a material batch are treated as representative of heterogeneous subsurface and hydrologic conditions.
3. **Model-terrain attack.** A model omits extreme events, blocked routes, erosion, settlement, vegetation change, maintenance failure, or adjacent-property effects.
4. **Passport-permit attack.** Complete material and site records are mistaken for engineering approval, land authority, water rights, or regulatory permission.
5. **Monitoring-rescue attack.** Post-deployment monitoring is used to justify inadequate pre-deployment evidence or an intervention without safe rollback.
6. **Data-custody attack.** Geologic, hydrologic, ecological, cultural, property, or infrastructure data is collected or disclosed beyond consent and purpose.
7. **Boundary-export attack.** Water, sediment, contamination, hazard, or maintenance burden is moved outside the measured site or onto another community.
8. **Climate-stationarity attack.** Historical conditions and design recurrence are treated as stable across changing climate and land use.

Claim and evidence ceiling

This brief specifies coupled investigation and decision modules. It does not establish site suitability, hydrologic benefit, material safety, structural stability, ecological improvement, legal authority, economic viability, or readiness to intervene. Each physical decision requires exact site and batch evidence, qualified engineering and environmental review, affected-party participation, applicable authority, safe failure and reversal planning, and continuing monitoring.

No site was assessed, material handled, structure designed, water routed, data released, external research performed, or implementation authorized in this pass.

Reader-facing consequence

The public value is a decision record that keeps material, site, model, authority, uncertainty, downstream exposure, monitoring, and revision together - including a clear result when the right answer is not to intervene.

Conclusion

The Hydroengineering and Geologic Intelligence Modules make Classified Earthworks more rigorous by coupling material transformation to site knowledge, intended use, evidence, authority, monitoring, and revision. Their most valuable product is not a universal landscape recipe. It is a governed method for asking what a material is, what evidence supports that description, where it may be used, what water will do there, who can decide, what happened after deployment, and when the answer must change.

A Classified Earthworks system does not merely ask the ground what it is. It records how the question was asked, who may rely on the answer, what remains unknown, and what consequences follow from acting on it.

Reference and Verification Note

Author corpus interfaces inspected under exact custody

Ableman Adam Ableman *Classified Earthworks.*

Ableman Adam Ableman *Technical Brief: Classified Earthworks System.*

External verification required

The source relies on or refers to EPA green-infrastructure and recharge material, NRCS soil-health guidance, ASTM D2487, USGS GeMS, FAIR principles, and CARE Principles for Indigenous Data Governance. No external source was consulted in this pass. Further evidentiary strengthening requires primary-source verification of wording, scope, current versions, applicability, and bibliographic details, plus qualified review of the scientific, engineering, legal, cultural, and governance claims.

Corpus-Wide Reference Positioning

This paper argues that the Hydroengineering and Geologic Intelligence Modules extend Classified Earthworks from material separation and transformation into a governed landscape-intervention architecture. Its nearest external comparison fields are earth, cps. The sources in [E1], [E2], [E3], [E4], [E5], [E6], [E7] are cited as predecessor literature, adjacent standards, or counterpressure - not as evidence that they caused the Ableman framework or that they validate its novel claims. The comparison isolates where established work already supplies provenance, formal assurance, institutional constraint, empirical method, or engineering

practice, and where this paper adds a distinct claim about admissibility, authority, semantic custody, or governed realization.

Within the Ableman corpus, the highest-priority current cross-links are [A1], [A2], [A3], [A4], [A5], [A6], [A7]. These are links among separately custodied works, not silent identity mergers. Their role is to expose mutual dependencies, terminology collisions, and places where a reader should traverse the corpus before treating this paper as self-sufficient.

Reference-pass counterevidence and limit search

The pass applied the following pressure tests: Material classes are sampling- and method-dependent; heterogeneous sites, contaminant transport, and regulatory thresholds can defeat a neat classification-to-action mapping. Runtime monitors and fallback controllers only protect modeled safety envelopes and can fail through sensor error, timing assumptions, unsafe recovery, or omitted hazards. These limits are not generic disclaimers: they identify the assumptions that a later empirical, formal, or implementation review must try to break.

Reference-pass influence and novelty boundary

No newly added external source is asserted to be a documented authorial influence unless the predecessor source already said so. `Predecessor` means earlier relevant work;

`adjacent` means a useful comparison; `standard` means an interoperability or assurance baseline; and `counterevidence` marks work that narrows or pressures a claim. Novelty therefore remains a reviewable, claim-level question rather than an inference from a low similarity score or an absent citation.

References

External scholarship, standards, and prior art

[**E1**] U.S. Army Corps of Engineers. *Sediment Investigations*, Engineer Manual EM 1110-2-4000. 14 February 2025. publications.usace.army.mil source **Relationship:** Primary engineering guidance for material characterization and classification; included to ground classification claims in field and laboratory practice.

[**E2**] U.S. Environmental Protection Agency. *Guidance for Cleaning Up Groundwater, Soil and Air at Hazardous Waste Cleanup Facilities.* epa.gov source **Relationship:** Primary remediation guidance emphasizing media transfer, exposure, and site-specific cleanup decisions.

[**E3**] Edward R. Griffor et al. *Framework for Cyber-Physical Systems: Volume 1, Overview.* NIST SP 1500-201, 2017. DOI: 10.6028/NIST.SP.1500-201. https://doi.org/10.6028/NIST.SP.1500-201 **Relationship:** Adjacent cyber-

physical systems framework for trustworthiness, timing, data, and lifecycle concerns.

[**E4**] W3C. *PROV-O: The PROV Ontology*. W3C Recommendation, 30 April 2013. https://www.w3.org/TR/prov-o/ **Relationship:** Adjacent provenance standard; it models entities, activities, and agents, but does not by itself establish the paper's stronger authority or standing claims.

[**E5**] J. Tanner Slagel et al. "A Formal Verification Framework for Runtime Assurance." *NASA Formal Methods*, 2024. NASA NTRS 20230017350. https://ntrs.nasa.gov/citations/20230017350 **Relationship:** Adjacent runtime-assurance formalization, useful for comparing failover safety with constitutional admission and effect recognition.

[**E6**] Nancy G. Leveson and John P. Thomas. *STPA Handbook*. MIT, 2018. https://psas.scripts.mit.edu/home/books-and-handbooks/ **Relationship:** Adjacent systems-theoretic hazard analysis method; used to test treatment of unsafe control and system-level constraints.

[**E7**] Robert N. M. Watson, Simon W. Moore, Peter Sewell, and Peter G. Neumann. *An Introduction to CHERI*. University of Cambridge Technical Report UCAM-CL-TR-941, 2019. DOI: 10.48456/tr-941. cl.cam.ac.uk source **Relationship:** Adjacent capability architecture demonstrating unforgeable hardware authority and compartmentalization.

Ableman corpus

[**A1**] Adam Ableman. *Classified Earthworks*. Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-8D3F34F396D7760E`;
source SHA-256
`9011C1800E2294A0F0E72BFBA871B071C8C4590C2A7E83FEBFF2BAE5D29E533A`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A2**] Adam Ableman. *Technical Brief: Classified Earthworks System*. Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-F4D74EA1DCD91309`;
source SHA-256
`3DCF2DFF9430B987F1D4BD8785D9A6ABB6309037FFC65ADA46F58AFD606DFFD8`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A3**] Adam Ableman. *Water: MCIR as the Engineering of Governed Change*. Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-91A910961BA1B5AF`;
source SHA-256
`1C0B814A76E0724FAD8ECC7A71DBBFC55B3C13E8ED03ED6F399496A70C2587CD`.

Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A4**] Adam Ableman. *Constitutional Robotics*. Ableman Corpus current edition; work identity `PAPER.PUB.18750683`;
source SHA-256
`D10DF9B6B7F742D9EEABF01BC5DFC3755BB4EBF548E8CB5F215BB9D94678F514`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A5**] Adam Ableman. *A Constitutional Syscall Architecture for Cyber-Physical Control Systems*. Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-F46D86D479F89BAC`;
source SHA-256
`5E696B753A10BD3C3D36B675A84B95CCF2234F024A6B277BF121E76BFCBCBD18`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A6**] Adam Ableman. *Effect Sovereignty*. Ableman Corpus current edition; work identity `PAPER.PUB.18737323`;
source SHA-256
`1DB8A34946A8705F6F4BA4FC3B12B251FC7C83DBDF217F9DDFF55529C8612EE7`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A7**] Adam Ableman. *Constitutionally Governed Infrastructure*. Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-EF5747107D56F516`;
source SHA-256
`9C708BA381108EEA252404DF33F7B3B1648EC9CF5F2ACF00BAB88C6A15540926`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

Editorial source notes retained from predecessor

The entries below are corpus-internal comparison and dependency sources. Their inclusion does not establish external validation of this paper's empirical, safety, novelty, or performance claims.

Adam Ableman *A Theory of Agentic Observation.* Zenodo, 2026. doi:10.5281/zenodo.20437813

Adam Ableman *CARA: Constrained Artificial Reasoning Architecture.* Zenodo, 2025. doi:10.5281/zenodo.18099715

Adam Ableman *Constitutional Computation.* Zenodo, 2026. doi:10.5281/zenodo.20624488

Adam Ableman *Constraint-Bearing Composability.* Zenodo, 2026. doi:10.5281/zenodo.21632061

Adam Ableman *Fractal Governance Architecture.* Zenodo, 2026. doi:10.5281/zenodo.20754189

Adam Ableman *Governed Semantic Compilation: Lawful Continuation, Refusal Semantics, and Epistemic Promotion in Large Language Models.* Zenodo, 2026. doi:10.5281/zenodo.18453681

Adam Ableman *MCIR: A Canonical Intermediate Representation for Governed Transformation.* Zenodo, 2026. doi:10.5281/zenodo.21095303

Adam Ableman *Semantic Coordinate Identity Tokenization (SCIT).* Zenodo, 2026. doi:10.5281/zenodo.20532070

Corpus source and companion formats: https://github.com/AblemanMosaic/ableman-corpus
License: CC-BY-ND-4.0

DOI: 10.5281/zenodo.23847969